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Environment	Kali Linux / Ubuntu	Submission	E-Campus

CORE102 LAB 3 REPORT

TASK 1 REPORT

EVIDENCE:

```
(kali@kali)~[~]
$ cd ~/Documents/ICDFA_LABS

(kali@kali)~[~/Documents/ICDFA_LABS]
$ mkdir core102_lab3

(kali@kali)~[~/Documents/ICDFA_LABS]
$ cd core102_lab3

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab3]
$ date
Fri Sep 25 06:08:30 AM EDT 2026

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab3]
$ hostname
kali

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab3]
$ ip -br addr
lo UNKNOWN 127.0.0.1/8 ::1/128
eth0 UP 192.168.149.128/24 fe80::96d8:5426:4bfc:2254/64
eth1 UP 192.168.113.128/24 fe80::e475:42ea:1db0:f1f7/64

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab3]
$ ip route
default via 192.168.149.2 dev eth0 proto dhcp src 192.168.149.128 metric 100
192.168.113.0/24 dev eth1 proto kernel scope link src 192.168.113.128 metric 101
192.168.149.0/24 dev eth0 proto kernel scope link src 192.168.149.128 metric 100

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab3]

Session Actions Edit View Help
192.168.113.0/24 dev eth1 proto kernel scope link src 192.168.113.128 metric 101
192.168.149.0/24 dev eth0 proto kernel scope link src 192.168.149.128 metric 100

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab3]
$ sudo tcpdump -D
[sudo] password for kali:
1.eth0 [Up, Running, Connected]
2.eth1 [Up, Running, Connected]
3.any (Pseudo-device that captures on all interfaces) [Up, Running]
4.lo [Up, Running, Loopback]
5.bluetooth-monitor (Bluetooth Linux Monitor) [Wireless]
6.nflog (Linux netfilter log (NFLOG) interface) [none]
7.nfqueue (Linux netfilter queue (NFQUEUE) interface) [none]
8.dbus-system (D-Bus system bus) [none]
9.dbus-session (D-Bus session bus) [none]

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab3]
$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            865M   0  865M   0% /dev
tmpfs           194M  1.3M  193M   1% /run
/dev/sda1       79G   73G   1.6G  98% /
tmpfs           968M  4.0K  968M   1% /dev/shm
none            1.0M   0   1.0M   0% /run/credentials/systemd-journald.service
tmpfs           968M  36K  968M   1% /tmp
none            1.0M   0   1.0M   0% /run/credentials/getty@tty1.service
tmpfs           194M  104K  194M   1% /run/user/1000
none            1.0M   0   1.0M   0% /run/credentials/systemd-networkd.service

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab3]
$
```

RESPONSE:

Capture	Interface	Capture filter	Expected packets	Stop condition
ICMP		icmp	Echo request and reply	10 packets
DNS		udp port 53 or tcp port 53	Query and response	20 packets
HTTP	lo	tcp port 8080	Handshake and HTTP request	After one request
SSH		tcp port 22	Handshake and encrypted session	After login and exit

In this Lab I used `sudo tcpdump -D` to check the interfaces available on my Kali so I can know which one to specify during the packet capturing and I also used a command `df -h` to check the disk space because the LAB recommended available 2GB space.

TASK 2 REPORT

EVIDENCE:

```
Session Actions Edit View Help
(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab3]
$ NET_IFACE=$(ip route show default | awk 'NR==1 {print $5}')
(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab3]
$ GATEWAY=$(ip route show default | awk 'NR==1 {print $3}')
(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab3]
$ echo "Interface: $NET_IFACE"
Interface: eth0
(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab3]
$ echo "Gateway: $GATEWAY"
Gateway: 192.168.149.2
(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab3]
$ sudo tcpdump -ni '$NET_IFACE' -c 10 -s0 -w core102_icmp.pcap
[sudo] password for kali:
tcpdump: $NET_IFACE: No such device exists
(No such device exists)
(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab3]
$ sudo tcpdump -ni '$NET_IFACE' -c 10 -s0 -w core102_icmp.pcap icmp
tcpdump: $NET_IFACE: No such device exists
(No such device exists)
(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab3]
$ sudo tcpdump -ni '$NET_IFACE' -c 10 -s0 -w core102_icmp.pcap icmp
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot len
^C8 packets captured
8 packets received by filter

PING 192.168.149.2 (192.168.149.2) 56(84) bytes of data.
64 bytes from 192.168.149.2: icmp_seq=1 ttl=128 time=0.549 ms
64 bytes from 192.168.149.2: icmp_seq=2 ttl=128 time=0.458 ms
64 bytes from 192.168.149.2: icmp_seq=3 ttl=128 time=0.604 ms
64 bytes from 192.168.149.2: icmp_seq=4 ttl=128 time=0.499 ms

— 192.168.149.2 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3161ms
rtt min/avg/max/mdev = 0.458/0.527/0.604/0.054 ms
(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab3]
$ NET_IFACE=$(ip route show default | awk 'NR==1 {print $5}')
(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab3]
$ ping -c 4 "$GATEWAY"

PING 192.168.149.2 (192.168.149.2) 56(84) bytes of data.
64 bytes from 192.168.149.2: icmp_seq=1 ttl=128 time=0.241 ms
64 bytes from 192.168.149.2: icmp_seq=2 ttl=128 time=0.766 ms
64 bytes from 192.168.149.2: icmp_seq=3 ttl=128 time=0.766 ms
64 bytes from 192.168.149.2: icmp_seq=4 ttl=128 time=0.486 ms

— 192.168.149.2 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3112ms
rtt min/avg/max/mdev = 0.241/0.564/0.766/0.219 ms
(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab3]
$
```

```
Session  Actions  Edit  View  Help
(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab3]
$ sudo tcpdump -ni "$NET_IFACE" -c 10 -s0 -w core102_icmp.pcap icmp
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
10 packets captured
10 packets received by filter
0 packets dropped by kernel

(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab3]
$ sudo tcpdump -nn -tttt -r core102_icmp.pcap
[sudo] password for kali:
reading from file core102_icmp.pcap, link-type EN10MB (Ethernet), snapshot length 262144
2026-09-25 08:31:21.463168 IP 192.168.149.128 > 192.168.149.2: ICMP echo request, id 2, seq 1, length 64
2026-09-25 08:31:21.463391 IP 192.168.149.2 > 192.168.149.128: ICMP echo reply, id 2, seq 1, length 64
2026-09-25 08:31:22.483643 IP 192.168.149.128 > 192.168.149.2: ICMP echo request, id 2, seq 2, length 64
2026-09-25 08:31:22.484362 IP 192.168.149.2 > 192.168.149.128: ICMP echo reply, id 2, seq 2, length 64
2026-09-25 08:31:23.490335 IP 192.168.149.128 > 192.168.149.2: ICMP echo request, id 2, seq 3, length 64
2026-09-25 08:31:23.491053 IP 192.168.149.2 > 192.168.149.128: ICMP echo reply, id 2, seq 3, length 64
2026-09-25 08:31:24.499138 IP 192.168.149.128 > 192.168.149.2: ICMP echo request, id 2, seq 4, length 64
2026-09-25 08:31:24.499576 IP 192.168.149.2 > 192.168.149.128: ICMP echo reply, id 2, seq 4, length 64
2026-09-25 08:40:42.995228 IP 192.168.149.128 > 192.168.149.254: ICMP
```

RESPONSE:

Frame	Source	Destination	ICMP type	Identifier or sequence	Meaning
1	192.168.149.128	192.168.149.2	ICMP	seq 1	Kali src (192.168.149.128) asks Gateway for ICMP request
	192.168.149.2	192.168.149.128	ICMP	seq 1	Gateway response
	192.168.149.128	192.168.149.2	ICMP	seq 2	Second response

Frame	Source	Destination	ICMP type	Identifier or sequence	Meaning
	192.168.149.2	192.168.149.128	ICMP	seq 2	Second response

In this Lab I used tcpdump on eth0 interface to capture icmp packets only and save it to core102_icmp.pcap file. Where I ping eth0 interface gateway and captured 10 packets. Icmp is all about confirming if the ip address or domain is reachable.

TASK 3 REPORT

EVIDENCE:

```
[sudo] password for kali:
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C4 packets captured
4 packets received by filter
0 packets dropped by kernel

(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab3]
$ sudo tcpdump -nn -vv -r core102_dns.pcap '(udp port 53 or tcp port 53)'
[sudo] password for kali:
reading from file core102_dns.pcap, link-type EN10MB (Ethernet), snapshot length 262144
10:01:26.600010 IP (tos 0x0, ttl 64, id 38771, offset 0, flags [none], proto UDP (17), length 80)
    192.168.149.128.35377 > 192.168.149.2.53: [bad udp cksum 0xac21 -> 0xb5d7!] 12058+ [1au] A? example.com. ar: . OPT UDPsize=1232 [COOKIE a9e0f3ea791ff8b8] (52)
10:01:26.632319 IP (tos 0x0, ttl 128, id 65421, offset 0, flags [none], proto UDP (17), length 100)
    192.168.149.2.53 > 192.168.149.128.35377: [udp sum ok] 12058 q: A? example.com. 2/0/1 example.com. A 104.20.23.154, example.com. A 172.66.147.243 ar: . OPT UDPsize=512 (72)
10:01:50.058272 IP (tos 0x0, ttl 64, id 11619, offset 0, flags [none], proto UDP (17), length 78)
    192.168.149.128.53282 > 192.168.149.2.53: [bad udp cksum 0xac1f -> 0xb6e!] 6566+ [1au] MX? gmail.com. ar: . OPT UDPsize=1232 [COOKIE 3f0562aba04fb6fb] (50)
10:01:50.091287 IP (tos 0x0, ttl 128, id 65423, offset 0, flags [none], proto UDP (17), length 189)
```

```

$ dig example.com A
dig gmail.com MX

; <<>> DiG 9.20.27-2-Debian <<>> example.com A
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 53981
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 512
;; QUESTION SECTION:
;example.com.                IN      A

;; ANSWER SECTION:
example.com.                 5       IN      A      172.66.147.243
example.com.                 5       IN      A      104.20.23.154

;; Query time: 44 msec
;; SERVER: 192.168.149.2#53(192.168.149.2) (UDP)
;; WHEN: Fri Sep 25 09:07:28 EDT 2026
;; MSG SIZE rcvd: 72

; <<>> DiG 9.20.27-2-Debian <<>> gmail.com MX
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 62701
;; flags: qr rd ra; QUERY: 1, ANSWER: 5, AUTHORITY: 0, ADDITIONAL: 1

```

```

; <<>> DiG 9.20.27-2-Debian <<>> gmail.com MX
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 62701
;; flags: qr rd ra; QUERY: 1, ANSWER: 5, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 512
;; QUESTION SECTION:
;gmail.com.                IN      MX

;; ANSWER SECTION:
gmail.com.                 5       IN      MX      20 alt2.gmail-smtp-in
.l.google.com.             5       IN      MX      40 alt4.gmail-smtp-in
gmail.com.                 5       IN      MX      5  gmail-smtp-in.l.goo
.l.google.com.             5       IN      MX      10 alt1.gmail-smtp-in
gmail.com.                 5       IN      MX      30 alt3.gmail-smtp-in
.l.google.com.

;; Query time: 24 msec
;; SERVER: 192.168.149.2#53(192.168.149.2) (UDP)
;; WHEN: Fri Sep 25 09:07:28 EDT 2026
;; MSG SIZE rcvd: 161

```

RESPONSE:

Query name	Type	Client port	Server	Response code	Answer summary
example.com	A	35377	53	NOERROR	It successfully searched and resolved the IPv4 associated to the domain "example.com"
gmail.com	MX	53282	53	NOERROR	It successfully queried the mail exchanger record for gmail.com. it returned the Gmail's mail server, showing where gmail.com is handled

In this LAB, I used tcpdump on eth0 to capture only dns packets by using dns port 53. dns checks which ip address is associated to a domain. When I dig example.com A, it resolved the ipv4 address associated to the domain “example.com” likewise when I “dig gmail.com XM”, it resolved and check the mail exchanger, that is, it checks the mail sever holding gmail.com.

TASK 4 REPORT

EVIDENCE:

```
Session Actions Edit View Help
127.0.0.1 - - [25/Sep/2026 11:50:22] "GET / HTTP/1.1" 200 -
^C
Keyboard interrupt received, exiting.

(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab3]
$ sudo tcpdump -nn -A -r core102_http.pcap 'tcp port 8080'
Stage Display filter Frame number Evidence observed

[sudo] password for kali:
tcpdump: truncated dump file; tried to read 4 file header bytes, only got 0
Stage: command not found

(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab3]
$ sudo tcpdump -nn -A -r core102_http.pcap 'tcp port 8080'
reading from file core102_http.pcap, link-type EN10MB (Ethernet), snapshot length 262144
11:50:22.278748 IP 127.0.0.1.46290 > 127.0.0.1.8080: Flags [S], seq 1901837801, win 65495, options [mss 65495,sackOK,TS val 1087452681 ecr 0,nop,wscale 8], length 0
E..4..@.@.B7...../...q[.8.
....(.....
@.6
@.6
11:50:22.278776 IP 127.0.0.1.8080 > 127.0.0.1.46290: Flags [S.], seq 797281591, ack 1901837802, win 65483, options [mss 65495,sackOK,TS val 1087452681 ecr 1087452681,nop,wscale 8], length 0
E..4..@.@.<...../..7q[.....0.....
@.6
@.6
11:50:22.278792 IP 127.0.0.1.46290 > 127.0.0.1.8080: Flags [.], ack 1, win 256, options [nop,nop,TS val 1087452681 ecr 1087452681], length 0
E..4..@.@./C.....q[../..8.....(.....

[nop,nop,TS val 1087452682 ecr 1087452682], length 0
E..4..@.@.B7...../...q[.8.
....(.....
@.6
@.6
11:50:22.280525 IP 127.0.0.1.46290 > 127.0.0.1.8080: Flags [F.], seq 79, ack 225, win 256, options [nop,nop,TS val 1087452682 ecr 1087452682], length 0
E..4..@.@./?......q[.8/....
....(.....
@.6
@.6
11:50:22.280557 IP 127.0.0.1.8080 > 127.0.0.1.46290: Flags [.], ack 80, win 256, options [nop,nop,TS val 1087452682 ecr 1087452682], length 0
E..4..@.@.B6...../...q[.9.
....(.....
@.6
@.6

(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab3]
$

* Trying 127.0.0.1:8080 ...
* Established connection to 127.0.0.1 (127.0.0.1 port 8080) from 127.0.0.1 port 46290
* using HTTP/1.x
> GET / HTTP/1.1
> Host: 127.0.0.1:8080
> User-Agent: curl/8.20.0
> Accept: */*
>
* Request completely sent off
* HTTP 1.0, assume close after body
< HTTP/1.0 200 OK
< Server: SimpleHTTP/0.6 Python/3.13.9
< Date: Fri, 25 Sep 2026 15:50:22 GMT
< Content-type: text/html
< Content-Length: 38
< Last-Modified: Fri, 25 Sep 2026 15:45:28 GMT
<
<h1>CORE102 authorised test page</h1>
* shutting down connection #0

(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab3]
$
```

```

$ sudo tcpdump -i eth0 -A -s 1 -w core102_http.pcap tcp port 8080
reading from file core102_http.pcap, link-type EN10MB (Ethernet), sn
apshot length 262144
11:50:22.278748 IP 127.0.0.1.46290 > 127.0.0.1.8080: Flags [S], seq
1901837801, win 65495, options [mss 65495,sackOK,TS val 1087452681 e
cr 0,nop,wscale 8], length 0
E..<..@.@./<.....q[.....0.....
@.6 .....
11:50:22.278776 IP 127.0.0.1.8080 > 127.0.0.1.46290: Flags [S.], seq
797281591, ack 1901837802, win 65483, options [mss 65495,sackOK,TS
val 1087452681 ecr 1087452681,nop,wscale 8], length 0
E..<..@.@.<...../..7q[.....0.....
@.6 @.6 ....
11:50:22.278792 IP 127.0.0.1.46290 > 127.0.0.1.8080: Flags [.], ack
1, win 256, options [nop,nop,TS val 1087452681 ecr 1087452681], leng
th 0
E..4..@.@./C.....q[../..8.....(.....
@.6 @.6 @.6 FileB.txt save
11:50:22.279058 IP 127.0.0.1.46290 > 127.0.0.1.8080: Flags [P.], seq
1:79, ack 1, win 256, options [nop,nop,TS val 1087452681 ecr 108745
2681], length 78: HTTP: GET / HTTP/1.1
E.....@.@.....q[../..8.....v.....
@.6 @.6 GET / HTTP/1.1
Host: 127.0.0.1:8080
User-Agent: curl/8.20.0
Accept: */*

```

```

1:79, ack 1, win 256, options [nop,nop,TS val 1087452681 ecr 108745
2681], length 78: HTTP: GET / HTTP/1.1
E.....@.@.....q[../..8.....v.....
@.6      @.6      GET / HTTP/1.1
Host: 127.0.0.1:8080
User-Agent: curl/8.20.0
Accept: */*

11:50:22.279067 IP 127.0.0.1.8080 > 127.0.0.1.46290: Flags [.], ack
79, win 256, options [nop,nop,TS val 1087452681 ecr 1087452681], len
gth 0
E..4..@.@.B:...../..8q[.8.....(.....
@.6      @.6
11:50:22.280290 IP 127.0.0.1.8080 > 127.0.0.1.46290: Flags [P.], seq
1:186, ack 79, win 256, options [nop,nop,TS val 1087452682 ecr 1087
452681], length 185: HTTP: HTTP/1.0 200 OK
E.....@.@.A...../..8q[.8.....
@.6
@.6      HTTP/1.0 200 OK
Server: SimpleHTTP/0.6 Python/3.13.9
Date: Fri, 25 Sep 2026 15:50:22 GMT
Content-type: text/html
Content-Length: 38
Last-Modified: Fri, 25 Sep 2026 15:45:28 GMT

11:50:22.280305 IP 127.0.0.1.46290 > 127.0.0.1.8080: Flags [.], ack
186, win 256, options [nop,nop,TS val 1087452682 ecr 1087452682], le
ngth 0
E..4..@.@./A.....q[.8/.....(.....

```

RESPONSE:

Stage	Display filter	Frame number	Evidence observed
Initial SYN	tcp.flags.syn == 1 && tcp.flags.ack == 0	1	127.0.0.1:46290 → 127.0.0.1:8080, Flags [S]
SYN ACK	tcp.flags.syn == 1 && tcp.flags.ack == 1	2	127.0.0.1:8080 → 127.0.0.1:46290, Flags [S.]
Final ACK	tcp.flags.ack == 1	3	Client sends ACK to complete the TCP three-way handshake

Stage	Display filter	Frame number	Evidence observed
HTTP request	http.request	4	GET / HTTP/1.1, Host 127.0.0.1:8080, User-Agent curl/8.20.0
HTTP response	http.response	6	Server sends HTTP/1.0 200 OK with Content-Type: text/html

In this LAB, I learned how to capture and analyze local HTTP traffic using tcpdump. I learned that a TCP connection begins with a **three-way handshake (SYN - SYN-ACK - ACK)** before data is exchanged. I also learned how to identify an **HTTP GET request**, an **HTTP 200 OK response**, and the **FIN/ACK packets** used to close the TCP connection. This helped me understand how a web request travels between a client and server, even when

TASK 5 REPORT

EVIDENCE:

The image displays two screenshots of the Wireshark network protocol analyzer interface. The top screenshot shows a packet capture of a TCP connection on port 8080. The bottom screenshot shows a packet capture of DNS traffic.

Top Screenshot: TCP Traffic (tcp.port == 8080)

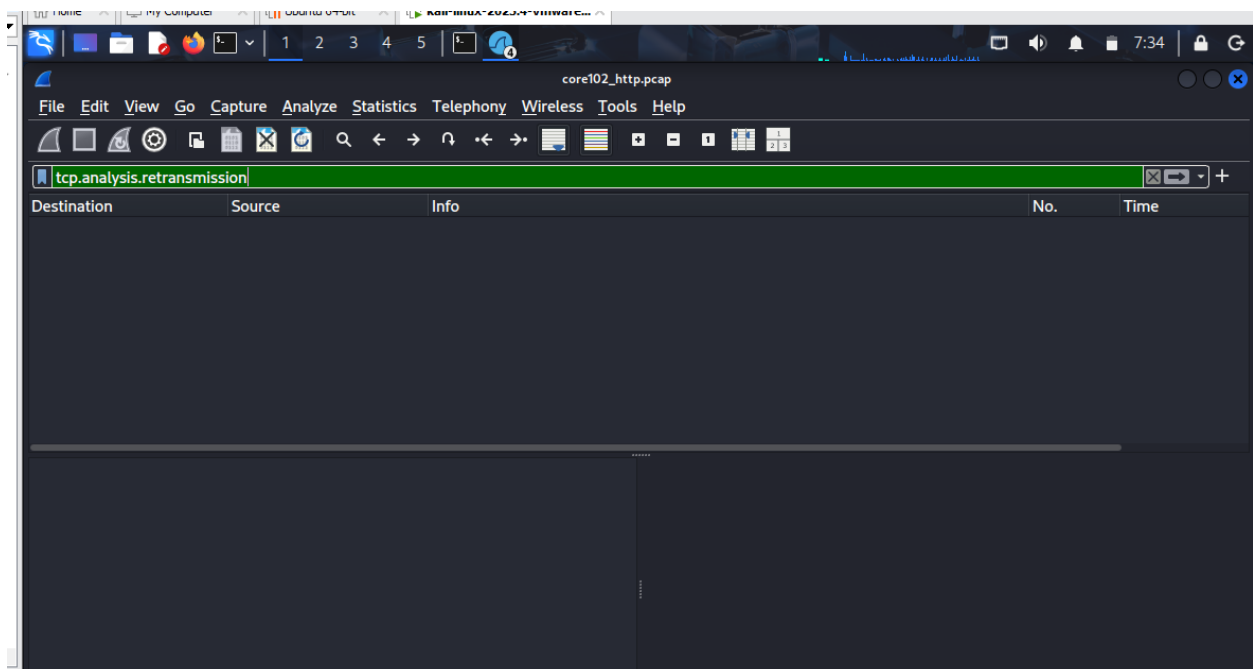
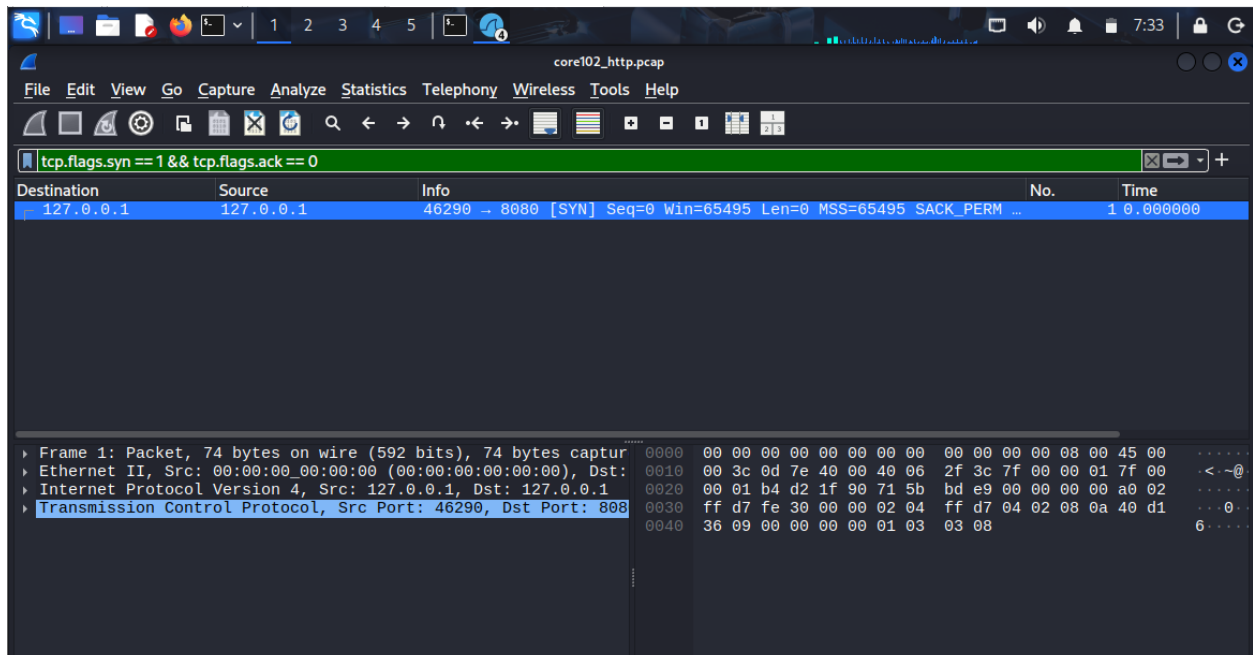
No.	Time	Destination	Source	Info
3	0.000044	127.0.0.1	127.0.0.1	46290 → 8080 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=10874526...
7	0.001557	127.0.0.1	127.0.0.1	46290 → 8080 [ACK] Seq=79 Ack=186 Win=65536 Len=0 TSval=10874...
9	0.001637	127.0.0.1	127.0.0.1	46290 → 8080 [ACK] Seq=79 Ack=224 Win=65536 Len=0 TSval=10874...
11	0.001777	127.0.0.1	127.0.0.1	46290 → 8080 [FIN, ACK] Seq=79 Ack=225 Win=65536 Len=0 TSval=...
5	0.000319	127.0.0.1	127.0.0.1	8080 → 46290 [ACK] Seq=1 Ack=79 Win=65536 Len=0 TSval=1087452...
12	0.001809	127.0.0.1	127.0.0.1	8080 → 46290 [ACK] Seq=225 Ack=80 Win=65536 Len=0 TSval=10874...
10	0.001682	127.0.0.1	127.0.0.1	8080 → 46290 [FIN, ACK] Seq=224 Ack=79 Win=65536 Len=0 TSval=...
6	0.001542	127.0.0.1	127.0.0.1	8080 → 46290 [PSH, ACK] Seq=1 Ack=79 Win=65536 Len=185 TSval=...
2	0.000028	127.0.0.1	127.0.0.1	8080 → 46290 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=65495...

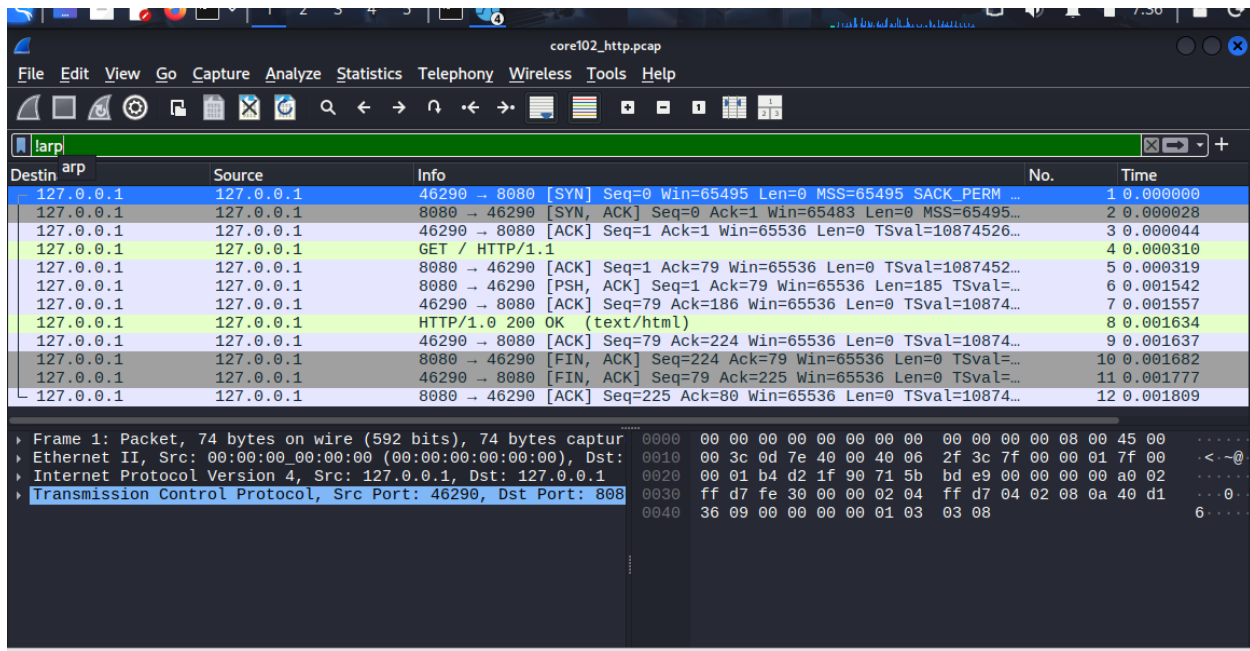
Frame 3: Packet, 66 bytes on wire (528 bits), 66 bytes captured on interface eth0, 66 bytes from 127.0.0.1:46290 to 127.0.0.1:8080 (00:00:00:00:00:00 to 00:00:00:00:00:00) [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=10874526...

Bottom Screenshot: DNS Traffic (dns)

No.	Time	Destination	Source	Info
1	0.000000	192.168.149.2	192.168.149.128	Standard query 0x2f1a A example.com OPT
2	0.032309	192.168.149.128	192.168.149.2	Standard query response 0x2f1a A example.com A 104.20.23.154 ...
3	23.458262	192.168.149.2	192.168.149.128	Standard query 0x19a6 MX gmail.com OPT
4	23.491277	192.168.149.128	192.168.149.2	Standard query response 0x19a6 MX gmail.com MX 5 gmail-smtp-i...

Frame 1: Packet, 94 bytes on wire (752 bits), 94 bytes captured on interface vnet0, 94 bytes from VMware_f7:72:d2 (00:0c:29:f7:72:d2) to 192.168.149.128 (08:00:27:1c:43:00) [Standard query] 0x2f1a A example.com OPT





RESPONSE:

Purpose	Wireshark display filter
Traffic to or from a host	ip.addr == 192.168.56.1
DNS traffic	dns
ICMP traffic	icmp
Local HTTP port	tcp.port == 8080
Initial TCP SYN packets	tcp.flags.syn == 1 && tcp.flags.ack == 0
TCP retransmissions	tcp.analysis.retransmission
Exclude ARP	!arp

Explain the difference between a capture filter and a display filter. Include one example of each and state when it is applied.

ANS: Capture filter **determines** what packets tcpdump/Wireshark captures in the first place e.g using tcpdump you can capture by running “sudo tcpdump -ni lo -w core102_http.pcap 'tcp port 8080' to capture tcp port 8080, you can use the GUI version of it called wireshark to that as well. However, both of them capture network traffic. **While A display filter** determines what packets Wireshark shows you after packets have already been captured. **Eg tcp.port == 8080**

TASK 6 REPORT

EVIDENCE:

```
Session Actions Edit View Help

System load: 1.07 Processes: 26
Usage of /home: unknown Users logged in: 0
Memory usage: 8% IPv4 address for eth0: 10.10.10.2
Swap usage: 0%

Expanded Security Maintenance for Applications is not enabled.

24 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Sat Sep 26 06:07:06 2026 from 192.168.113.128
dyahayaaddra11@ubuntu:~$ hostname
ubuntu
dyahayaaddra11@ubuntu:~$ whoami
dyahayaaddra11
dyahayaaddra11@ubuntu:~$ ls
core102_lab2
dyahayaaddra11@ubuntu:~$ exit
logout
Connection to 192.168.113.129 closed.

(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab3]
$ sudo tcpdump -ni eth1 -s0 -w core102_ssh.pcap 'tcp port 22'
tcpdump: listening on eth1, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C1723 packets captured
1723 packets received by filter
0 packets dropped by kernel

(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab3]
$ sudo tcpdump -nn -r core102_ssh.pcap 'tcp port 22'
reading from file core102_ssh.pcap, link-type EN10MB (Ethernet), snapshot length 262144
02:12:02.355965 IP 192.168.113.128.54548 > 192.168.113.129.22: Flags [S], seq 3051719624, win 64240, options [mss 1460,sackOK,TS val 666275713 ecr 0,nop,wscale 8], length 0
02:12:02.356977 IP 192.168.113.129.22 > 192.168.113.128.54548: Flags [S.], seq 120015349, ack 3051719625, win 65160, options [mss 1460,sackOK,TS val 3049860868 ecr 666275713,nop,wscale 8], length 0
02:12:02.357087 IP 192.168.113.128.54548 > 192.168.113.129.22: Flags [P.], seq 1, ack 1, win 251, options [nop,nop,TS val 666275715 ecr 3049860868], length 33: SSH: SSH-2.0-OpenSSH_10.2p1 Debian-2
02:12:02.359026 IP 192.168.113.129.22 > 192.168.113.128.54548: Flags [P.], seq 34, win 255, options [nop,nop,TS val 3049860870 ecr 666275715], length 0
02:12:02.522707 IP 192.168.113.129.22 > 192.168.113.128.54548: Flags
```

RESPONSE:

In this LAB, I used tcpdump to capture ssh packets. I ssh through my Kalli and listen to the ssh packets on the other terminal on my kali and after I ran some command on my ssh such as hostname, whoami, and finally exit the ssh and went back to my second terminal and do ctrl C to stop the listening and I captured almost 1000 packets on the ssh port. I got to know that when I read my captured file.

TASK 7 REPORT

EVIDENCE:

RESPONSE:

Time	Frame	Source	Destination	Protocol	Event and interpretation
02:12:02.355965	1	192.168.113.128:54548	192.168.113.129:22	TCP	SYN Kali initiates an SSH TCP connection to Ubuntu on port 22. This is the first communication attempt.
02:12:02.356977	2	192.168.113.129:22	192.168.113.128:54548	TCP	SYN-ACK Ubuntu acknowledges the connection request and indicates that the SSH port is available.
02:12:02.357087	3	192.168.113.128:54548	192.168.113.129:22	TCP	ACK Kali acknowledges Ubuntu's response, completing the TCP three-way handshake.
02:12:02.357993	4	192.168.113.128:54548	192.168.113.129:22	SSH	SSH request/version identification Kali sends SSH-2.0-OpenSSH_10.2p1 Debian-2 to identify its SSH implementation.
02:12:02.522707	6	192.168.113.129:22	192.168.113.128:54548	SSH	Matching response Ubuntu responds with SSH-2.0-OpenSSH_10.2p1

Time	Frame	Source	Destination	Protocol	Event and interpretation
					Ubuntu-2ubuntu3.6, confirming its SSH implementation.
02:12:02.523525	8	192.168.113.128:54548	192.168.113.129:22	SSH	Encrypted SSH session data begins. The payload is not displayed as readable commands because SSH encrypts the session.
02:12:02.555015	10	192.168.113.129:22	192.168.113.128:54548	SSH	Ubuntu sends additional SSH session data. TCP sequence and acknowledgment numbers maintain reliable communication.
02:12:02.559617	11	192.168.113.128:54548	192.168.113.129:22	SSH	Kali sends additional encrypted SSH session data.
02:12:02.569926	12	192.168.113.129:22	192.168.113.128:54548	SSH	Ubuntu responds with additional SSH session data. The SSH conversation continues normally.